Generalizable Metric Network for Cross-Domain Person Re-Identification

Abstract:
Person Re-Identification (Person Re-ID) across non-overlapping camera networks suffers severe performance d

1. Introduction & Motivation:
Current deep learning Re-ID models achieve high Rank-1 accuracy when trained and tested on the same datase

2. Proposed Methodology (GMN Architecture):
GMN consists of three primary modules:
(1) Feature Extractor Backbone: ResNet-50 initialized with ImageNet pre-trained weights.
(2) Domain-Adversarial Module: A domain discriminator that penalizes domain-specific feature cluster formations
(3) Meta-Metric Loss (MML): A meta-learning loss function using Episodic Source Domain Splitting to simulate d

3. Experiments & Results:
We benchmark GMN across Market-1501 -> DukeMTMC-reID and Market-1501 -> MSMT17.
- Market-1501 to DukeMTMC-reID: Rank-1 Accuracy = 78.4%, mAP = 62.1% (outperforming baseline ResNet-50
- Market-1501 to MSMT17: Rank-1 Accuracy = 44.8%, mAP = 28.5%.

4. Limitations & Future Work:
While GMN shows strong generalization across supervised source domains, it requires labeled identity annotatio